Logarithms as Inverses of Exponential Functions

Pre-calculus 12 — Unit: Logarithmic Functions and Equations

Lesson 4 of 5 • 90-Minute Period

Lesson Agenda

Time	Activity	Focus
0–5 min	Warm-up review	Exponential functions & inverses
5–20 min	Concept development	What does $\log_b x$ mean?
20–35 min	Graphical exploration	$y = b^x$ and $y = \log_b x$ as reflections
35–50 min	Worked examples I	Converting, evaluating, sketching
50–60 min	Think-pair-share	Key properties & misconceptions
60–75 min	Worked examples II	Harder problems, domain, range
75–85 min	Student practice	Independent work
85–90 min	Summary & exit card	Consolidation

Learning Goals

By the end of this lesson you will be able to:

- explain **why** a logarithm is the inverse of an exponential
- convert fluently between **exponential** and **logarithmic** form
- **sketch** the graph of $y = \log_b x$ from the graph of $y = b^x$, and identify key features
- state and apply the **domain, range, and asymptote** of a logarithmic function
- verify that f and g are inverses using the **composition cancellation** properties

Prerequisite check

You should already be comfortable with: laws of exponents, graphing $y = b^x$, the definition of an inverse function, and reflecting graphs across $y = x$.

Warm-Up (5 min) — Solve for the Exponent

Each question asks: *to what power must I raise the base to get the result?*

1. $2^? = 8$
2. $3^? = 81$
3. $5^? = 1$
4. $10^? = 10\,000$
5. $4^? = \frac{1}{16}$
6. $b^? = b$
7. $b^? = 1$
8. $2^? = \sqrt{2}$

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Answers: (1) 3 (2) 4 (3) 0 (4) 4 (5) -2 (6) 1 (7) 0 (8) $\frac{1}{2}$

Each answer *is* a logarithm — you have been computing them intuitively. Today we give them a name and notation.

The Core Question Exponentials Cannot Answer Directly

Exponential form answers:

"What is b raised to a given power?"

$$b^x = y$$

Example: $2^5 = 32$

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"What power of b gives y?"

$$\log_b y = x$$

Example: $\log_2 32 = 5$

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Key relationship (learn this cold)

$$b^x = y \iff \log_b y = x \qquad (b > 0,\ b \neq 1,\ y > 0)$$

The logarithm **is the exponent**. Nothing more, nothing less.

Formal Definition of a Logarithm

Definition

For $b > 0$, $b \neq 1$, and $y > 0$:

$$\log_b y = x \quad \text{means} \quad b^x = y$$

Read: *"the logarithm base b of y equals x."*

Anatomy of logarithmic notation:

logarithm symbol (must be exponent)

argument (must be > 0)

$$\log_b \quad y \quad = \quad x$$

base

Example

$\log_3 27 = 3$ because $3^3 = 27$

Conversion: Exponential $\leftrightarrow$ Logarithmic Form

Exponential $\to$ Log

$4^3 = 64 \quad \Rightarrow \quad \log_4 64 = 3$

$7^0 = 1 \quad \Rightarrow \quad \log_7 1 = 0$

$2^{-3} = \frac{1}{8} \quad \Rightarrow \quad \log_2 \frac{1}{8} = -3$

$e^1 = e \quad \Rightarrow \quad \ln e = 1$

$10^4 = 10\,000 \quad \Rightarrow \quad \log 10\,000 = 4$

Log $\to$ Exponential

$\log_6 36 = 2 \quad \Rightarrow \quad 6^2 = 36$

$\log_b 1 = 0 \quad \Rightarrow \quad b^0 = 1$

$\log_3 \frac{1}{9} = -2 \quad \Rightarrow \quad 3^{-2} = \frac{1}{9}$

$\ln x = 5 \quad \Rightarrow \quad e^5 = x$

$\log k = 3 \quad \Rightarrow \quad 10^3 = k$

Special notation: $\log x$ (no base written) means $\log_{10} x$ — the **common logarithm**.
$\ln x$ means $\log_e x$ — the **natural logarithm**, base $e \approx 2.718$.

Logarithm as the Inverse Function

Recall: to find the inverse of f, swap x and y and solve for y.

Start with:

$$f(x) = b^x$$

Step 1 — write as $y = b^x$.
Step 2 — swap: $x = b^y$.
Step 3 — solve for y:

$$y = \log_b x$$

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Therefore

$$f(x) = b^x \implies f^{-1}(x) = \log_b x$$

The logarithmic function is **exactly** the inverse of the exponential function with the same base.

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Cancellation (composition) properties

$$\log_b(b^x) = x \quad \text{for all } x \in \mathbb{R}$$

$$b^{\log_b x} = x \quad \text{for all } x > 0$$

These are the "undo" properties — applying f then f^{-1} (or vice versa) returns the original input.

Cancellation Properties in Action

Use the cancellation properties to simplify **without a calculator**:

$\log_b(b^x) = x$

$$\log_3(3^7) = 7$$

$$\log_5(5^{-2}) = -2$$

$$\log(10^{4.5}) = 4.5$$

$$\ln(e^{2t}) = 2t$$

$b^{\log_b x} = x$

$$3^{\log_3 11} = 11$$

$$7^{\log_7 0.4} = 0.4$$

$$10^{\log \sqrt{2}} = \sqrt{2}$$

$$e^{\ln(3x)} = 3x \quad (x > 0)$$

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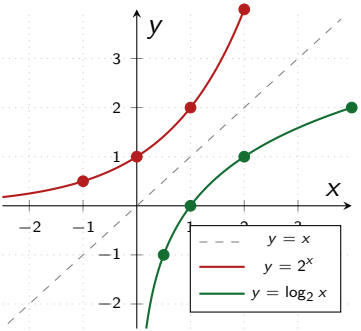
Why does this work? $\log_b(b^x)$ asks "to what power do I raise b to get b^x?" The answer is obviously x. Similarly, $b^{\log_b x}$ raises b to the power that produces x — giving x back.

Graphing $y = \log_b x$ from $y = b^x$

Since f^{-1} is obtained by reflecting across $y = x$:



Key observations:

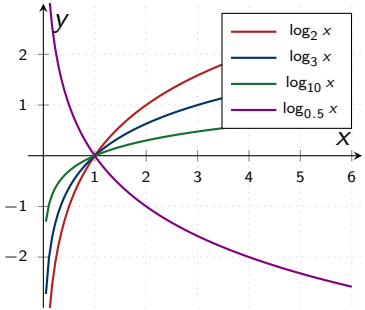
- The two curves are **reflections** of each other across $y = x$.
- Every point (a, b) on $y = 2^x$ has a **mirror point** (b, a) on $y = \log_2 x$.
- The x-axis is a horizontal asymptote of $y = 2^x$; the y-axis ($x = 0$) is a vertical asymptote of $y = \log_2 x$.
- Both curves pass through $(1, 0) \leftrightarrow (0, 1)$.

Key Features: $y = b^x$ vs $y = \log_b x$

Feature	$y = b^x$	$y = \log_b x$
Domain	$(-\infty, \infty)$	$(0, \infty)$
Range	$(0, \infty)$	$(-\infty, \infty)$
x-intercept	none	$(1, 0)$
y-intercept	$(0, 1)$	none
Asymptote	$y = 0$ (horizontal)	$x = 0$ (vertical)
Passes through	$(1, b)$	$(b, 1)$
Behaviour ($b > 1$)	increasing, concave up	increasing, concave down
Behaviour ($0 < b < 1$)	decreasing, concave up	decreasing, concave down

Domain restriction — the most important consequence

Effect of the Base on the Graph



All $y = \log_b x$ graphs:

- Pass through $(1, 0)$
- Have $x = 0$ as vertical asymptote
- Have domain $(0, \infty)$, range $(-\infty, \infty)$

Base determines slope and direction:

- $b > 1$: **increasing** (larger base $\Rightarrow$ flatter curve)
- $0 < b < 1$: **decreasing**
- All curves pass through $(b, 1)$

The x-intercept is **always** $(1, 0)$ because $\log_b 1 = 0$ for any valid base.

Worked Example 1 — Evaluating Logarithms

Evaluate each logarithm **without a calculator**.

(a) $\log_4 64$

Ask: $4^? = 64$

$4^1 = 4,\ \ 4^2 = 16,\ \ 4^3 = 64$ ✓

$$\log_4 64 = \mathbf{3}$$

(b) $\log_9 3$

Ask: $9^? = 3$

$9^{1/2} = \sqrt{9} = 3$ ✓

$$\log_9 3 = \mathbf{\tfrac{1}{2}}$$

(c) $\log_2 \dfrac{1}{32}$

Ask: $2^? = \frac{1}{32} = 2^{-5}$

$$\log_2 \tfrac{1}{32} = \mathbf{-5}$$

(d) $\log_7 1$

Ask: $7^? = 1$

$7^0 = 1$ for any base ✓

$$\log_7 1 = \mathbf{0}$$

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Ask: $4^? = 64$
$4^1 = 4,\ \ 4^2 = 16,\ \ 4^3 = 64$ ✓

$$\log_4 64 = \mathbf{3}$$

(b) $\log_9 3$

Ask: $9^? = 3$
$9^{1/2} = \sqrt{9} = 3$ ✓

$$\log_9 3 = \mathbf{\tfrac{1}{2}}$$

(c) $\log_2 \dfrac{1}{32}$

Ask: $2^? = \frac{1}{32} = 2^{-5}$

$$\log_2 \tfrac{1}{32} = \mathbf{-5}$$

(d) $\log_7 1$

Ask: $7^? = 1$
$7^0 = 1$ for any base ✓

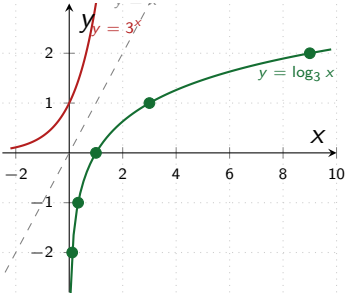
$$\log_7 1 = \mathbf{0}$$

Strategy: always rewrite the question as "b to the power __ equals the argument," then find that power.

Worked Example 2 — Sketching $y = \log_3 x$

Step 1: Build a table using the inverse relationship.

$y = 3^x$		$y = \log_3 x$	
x	y	x	y
-2	$\frac{1}{9}$	$\frac{1}{9}$	-2
-1	$\frac{1}{3}$	$\frac{1}{3}$	-1
0	1	1	0
1	3	3	1
2	9	9	2



Step 2: Plot the swapped coordinates. **Step 3:** Join with a smooth curve. The vertical asymptote is $x = 0$.

Worked Example 3 — Verifying Inverse Functions

Show that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses.

Two functions are inverses iff $f(g(x)) = x$ **and** $g(f(x)) = x$.

Compute $f(g(x))$

$$\begin{aligned} f(g(x)) &= f(\log_2 x) \\ &= 2^{\log_2 x} \\ &= x \quad \checkmark \end{aligned}$$

(by the cancellation property $b^{\log_b x} = x$, valid for $x > 0$)

Compute $g(f(x))$

$$\begin{aligned} g(f(x)) &= g(2^x) \\ &= \log_2(2^x) \\ &= x \quad \checkmark \end{aligned}$$

(by the cancellation property $\log_b(b^x) = x$, valid for all $x \in \mathbb{R}$)

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Conclusion

Since $f(g(x)) = x$ for all $x > 0$ and $g(f(x)) = x$ for all $x \in \mathbb{R}$, we have confirmed f and g are

Think-Pair-Share — Spot the Error

A student wrote the following. What is wrong?

"Since $f(x) = 3^x$ has domain $(-\infty, \infty)$ and range $(0, \infty)$, its inverse $f^{-1}(x) = \log_3 x$ also has domain $(-\infty, \infty)$ and range $(0, \infty)$. Therefore $\log_3(-5)$ is defined and equals $-\log_3 5$."

Discuss with a partner for 2 minutes, then share.

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Discuss with a partner for 2 minutes, then share.

Correction

When you take the inverse, **domain and range swap**:

$$f : \text{dom } (-\infty, \infty) \to \text{range } (0, \infty) \quad \Longrightarrow \quad f^{-1} : \text{dom } (0, \infty) \to \text{range } (-\infty, \infty)$$

So $\log_3 x$ has domain $(0, \infty)$. $\log_3(-5)$ is **undefined** because $-5 < 0$ — no power of 3 can produce a negative number.
Also, $-\log_3 5 = \log_3 5^{-1} = \log_3 \frac{1}{5}$, which is a different value entirely.

Worked Example 4 — Domain of a Logarithmic Function

Find the domain of each function. **Argument must be strictly positive.**

(a) $f(x) = \log_4(x-3)$

Require: $x - 3 > 0$

$$x > 3 \quad \Longrightarrow \quad \text{Domain: } (3, \infty)$$

Vertical asymptote: $x = 3$

(b) $f(x) = \log_2(5-2x)$

Require: $5 - 2x > 0$
$-2x > -5 \Rightarrow x < \frac{5}{2}$

$$\text{Domain: } \left(-\infty, \frac{5}{2}\right)$$

(c) $f(x) = \ln(x^2 - 9)$

Require: $x^2 - 9 > 0$
$(x-3)(x+3) > 0$
Sign chart:

	$x < -3$	$-3 < x < 3$	$x > 3$
$(x-3)$	$-$	$-$	$+$
$(x+3)$	$-$	$+$	$+$
Product	$+$	$-$	$+$

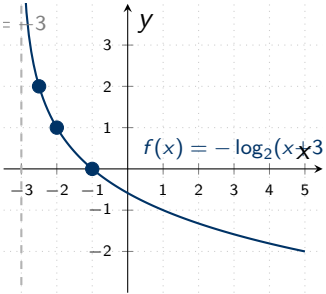
$$\text{Domain: } (-\infty, -3) \cup (3, \infty)$$

Worked Example 5 — Transformations of $y = \log_b x$

Sketch $f(x) = -\log_2(x+3) + 1$. State domain, range, and asymptote.

Identify the transformations from $y = \log_2 x$:

- $(x+3)$: horizontal shift **left 3**
- $-\log_2$: reflection in the x-**axis**
- $+1$: vertical shift **up 1**



Key reference points on $y = \log_2 x$:
$(1,0)$, $(2,1)$, $(\frac{1}{2}, -1)$

Mapped points (shift left 3, reflect, shift up 1): $(1,0) \to (-2,1)$, $(2,1) \to (-1,0)$, $(\frac{1}{2}, -1) \to (-\frac{5}{2}, 2)$

Vertical asymptote: $x = -3$

Domain: $(-3, \infty)$

Worked Example 6 — Finding an Inverse Algebraically

Find the inverse of $f(x) = \log_3(2x - 1)$ and state its domain and range.

Step 1: Replace $f(x)$ with y.

$$y = \log_3(2x - 1)$$

Step 2: Swap x and y.

$$x = \log_3(2y - 1)$$

Step 3: Solve for y — convert to exponential form.

$$3^x = 2y - 1$$

$$2y = 3^x + 1$$

$$y = \frac{3^x + 1}{2}$$

Domain and range

For $f(x) = \log_3(2x - 1)$:

- Require $2x - 1 > 0 \Rightarrow x > \frac{1}{2}$
- Domain of f: $\left(\frac{1}{2}, \infty\right)$
- Range of f: $(-\infty, \infty)$

For $f^{-1}(x) = \frac{3^x+1}{2}$:

- Domain = Range of f: $(-\infty, \infty)$
- Range = Domain of f: $\left(\frac{1}{2}, \infty\right)$

Verify: $f^{-1}(f(x)) = \frac{3^{\log_3(2x-1)} + 1}{2}$

Common Misconceptions

Misconception	Correction
$\log_b(-x)$ **is defined and equals** $-\log_b x$**.**	$\log_b(-x)$ is **undefined** for $x > 0$; there is no real power of b that gives a negative number.
$\log_b(x+y) = \log_b x + \log_b y$	The product law says $\log_b(xy) = \log_b x + \log_b y$. Adding inside a log is **not** the same as multiplying.
log **means** $\log_2$	log with no base means $\log_{10}$ (common log). In means $\log_e$ (natural log).
Domain of $\log_b x$ **is all reals.**	Domain is $(0, \infty)$ only. Always verify the argument is **strictly positive**.
$\log_b x$ **is the reciprocal of** b^x	Logarithm is the **inverse function**, not the reciprocal. $\frac{1}{b^x} = b^{-x}$, not $\log_b x$.

Student Practice (10 min)

Work on the following independently, then compare with a partner.

1. Evaluate without a calculator: $\log_5 125$, $\log_4 \frac{1}{64}$, $\log_8 2$.
2. Convert to logarithmic form: $6^3 = 216$, $e^k = 15$.
3. Convert to exponential form: $\log_2 32 = 5$, $\ln 7 = t$.
4. Simplify: $4^{\log_4 9}$, $\log_3(3^{2x+1})$.
5. Find the domain of $f(x) = \log_5(4 - x^2)$.
6. Find the inverse of $f(x) = 5^{3x-2}$ and state the domain and range of the inverse.
7. Sketch $g(x) = \log_2(-x) + 3$. State the domain, range, asymptote, and one point on the graph.

Practice Solutions

1. $\log_5 125 = 3$; $\log_4 \frac{1}{64} = -3$; $\log_8 2 = \frac{1}{3}$.

2. $6^3 = 216 \Rightarrow \log_6 216 = 3$;
$e^k = 15 \Rightarrow \ln 15 = k$.

3. $\log_2 32 = 5 \Rightarrow 2^5 = 32$; $\ln 7 = t \Rightarrow e^t = 7$.

4. $4^{\log_4 9} = 9$; $\log_3(3^{2x+1}) = 2x + 1$.

5. Require
$4 - x^2 > 0 \Rightarrow x^2 < 4 \Rightarrow -2 < x < 2$.
Domain: $(-2, 2)$.

6. $y = 5^{3x-2}$; swap: $x = 5^{3y-2}$; take $\log_5$:
$\log_5 x = 3y - 2$; solve: $y = \frac{\log_5 x + 2}{3}$.
$f^{-1}(x) = \dfrac{\log_5 x + 2}{3}$, domain $(0, \infty)$, range $(-\infty, \infty)$.

7. $g(x) = \log_2(-x) + 3$; need
$-x > 0 \Rightarrow x < 0$.
Domain: $(-\infty, 0)$; range: $(-\infty, \infty)$;
asymptote: $x = 0$; passes through $(-1, 3)$.

Lesson Summary

The big ideas from today

1. $b^x = y \iff \log_b y = x$. The log **is** the exponent.
2. $f(x) = b^x$ and $g(x) = \log_b x$ are **inverse functions**.
3. Their graphs are **reflections** of each other across $y = x$.
4. Swapping gives swapped domain/range: b^x has dom $\mathbb{R}$, range $(0, \infty)$; $\log_b x$ has dom $(0, \infty)$, range $\mathbb{R}$.
5. Cancellation: $\log_b(b^x) = x$ and $b^{\log_b x} = x$.
6. Always check that the **argument** of a log is > 0.

Coming next (Lesson 5):

- Solving logarithmic equations
- Situational (applied) problems
- Extraneous roots

Connections to prior lessons:

- Lessons 1–3: laws of logs, evaluating, graphing
- Transformations unit: $y = af(b(x - h)) + k$

Exit Card

Answer on a piece of paper before you leave.

Question

Let $f(x) = 4^x$.

a. Write $f^{-1}(x)$.
b. State the domain and range of f^{-1}.
c. Verify that $f(f^{-1}(x)) = x$ using the cancellation property.
d. Evaluate $f^{-1}(64)$ without a calculator. Show your reasoning.

Answer

(a) $f^{-1}(x) = \log_4 x$. (b) Domain $(0, \infty)$, range $(-\infty, \infty)$.
(c) $f(f^{-1}(x)) = 4^{\log_4 x} = x$ ✓ (d) $\log_4 64$: need $4^? = 64 = 4^3$, so $\log_4 64 = 3$.